

Raven and Johnson: Biology - Photosynthesis

In photosynthesis, autotrophs capture light energy and store it as chemical energy in sugars. The reactions take place inside the chloroplast.

The light reactions occur in the thylakoid membranes. They generate ATP and NADPH, split water, and release oxygen.

The Calvin cycle takes place in the stroma and proceeds in three phases: carbon fixation, reduction, and regeneration of the acceptor molecule. The enzyme rubisco catalyses the carbon fixation phase.

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During carbon fixation, rubisco attaches carbon dioxide to a five-carbon acceptor molecule, beginning the Calvin cycle.

The rate of photosynthesis increases with light intensity until it levels off at the light saturation point, beyond which more light does not help.